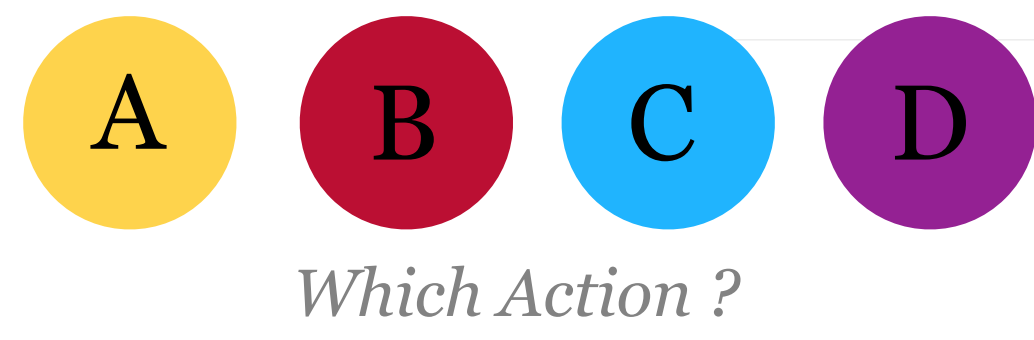


Introduction & Background

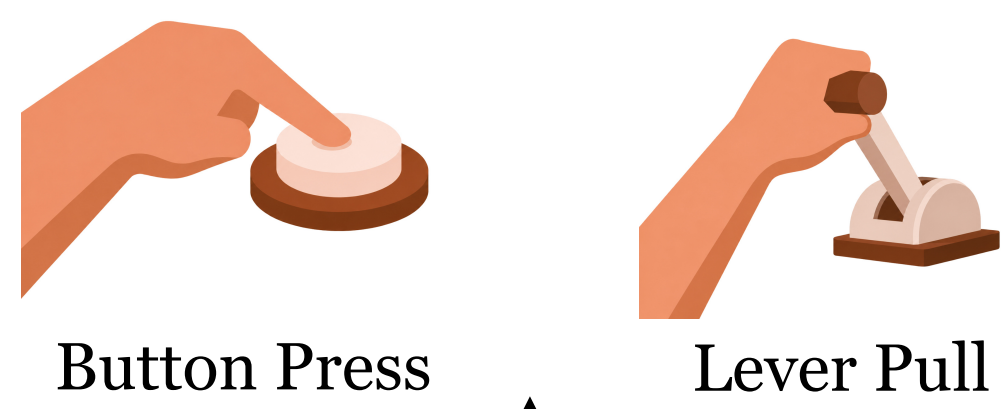
1 Motor Control: Two Coupled Problems

Action Selection

Discrete action choice



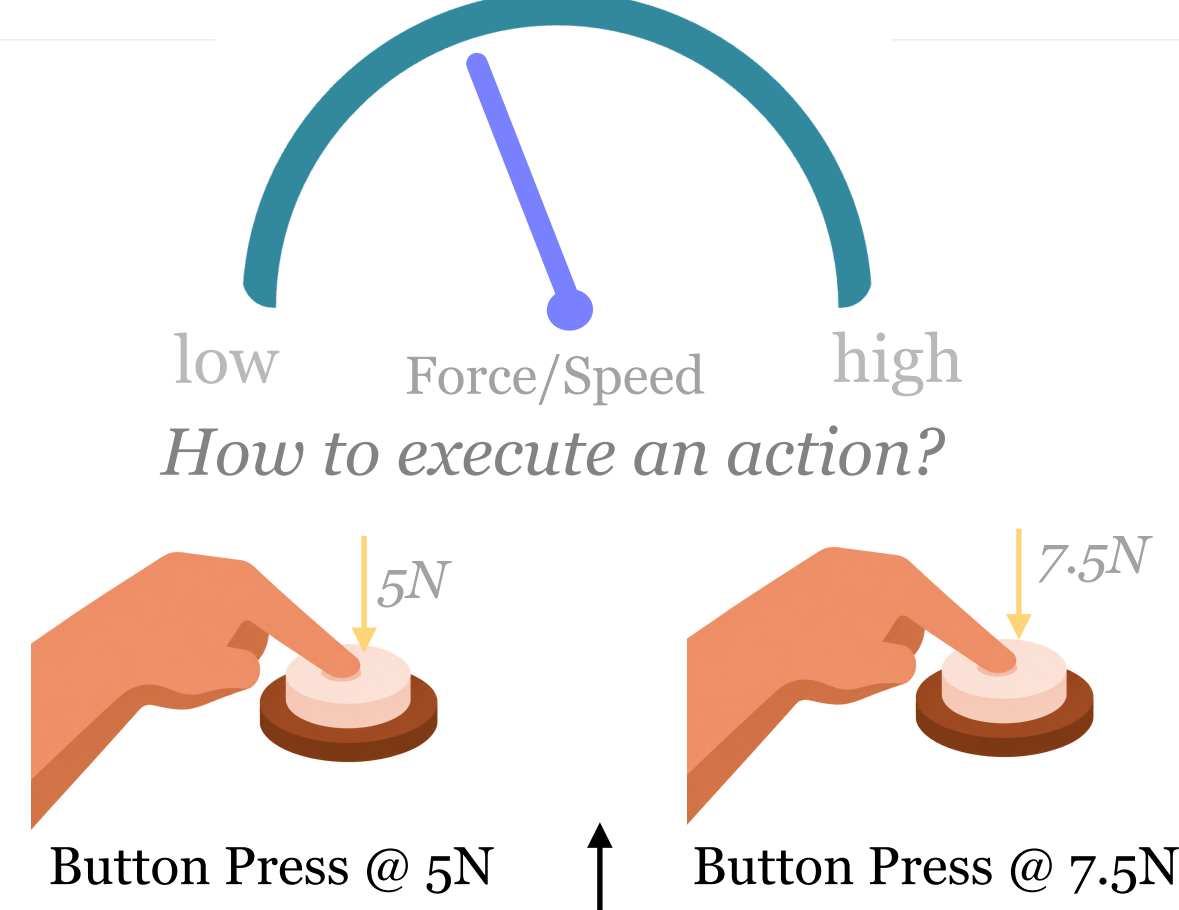
Which Action?



Basal Ganglia
Selects the discrete action

Action Specification

Continuous Action Parameter



Motor Cortex
Specifies the action execution

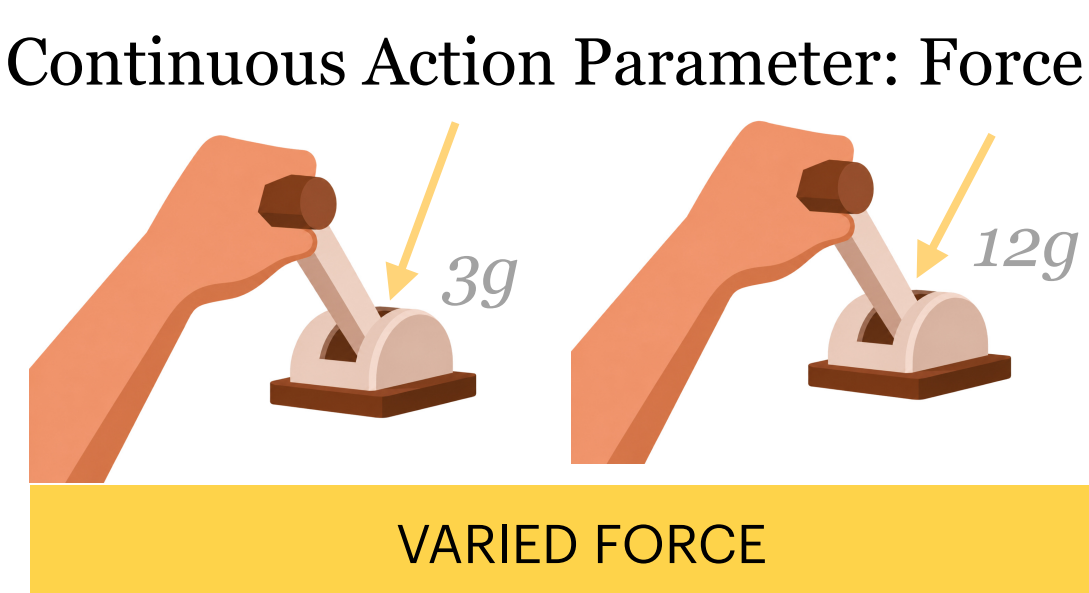
2 Recent Evidence: Basal Ganglia also specifies!

*Park et al., 2025

- Systematically varied the continuous action parameter while keeping the gross action constant
- Allowed to test if the basal ganglia also specifies continuous kinematic details.

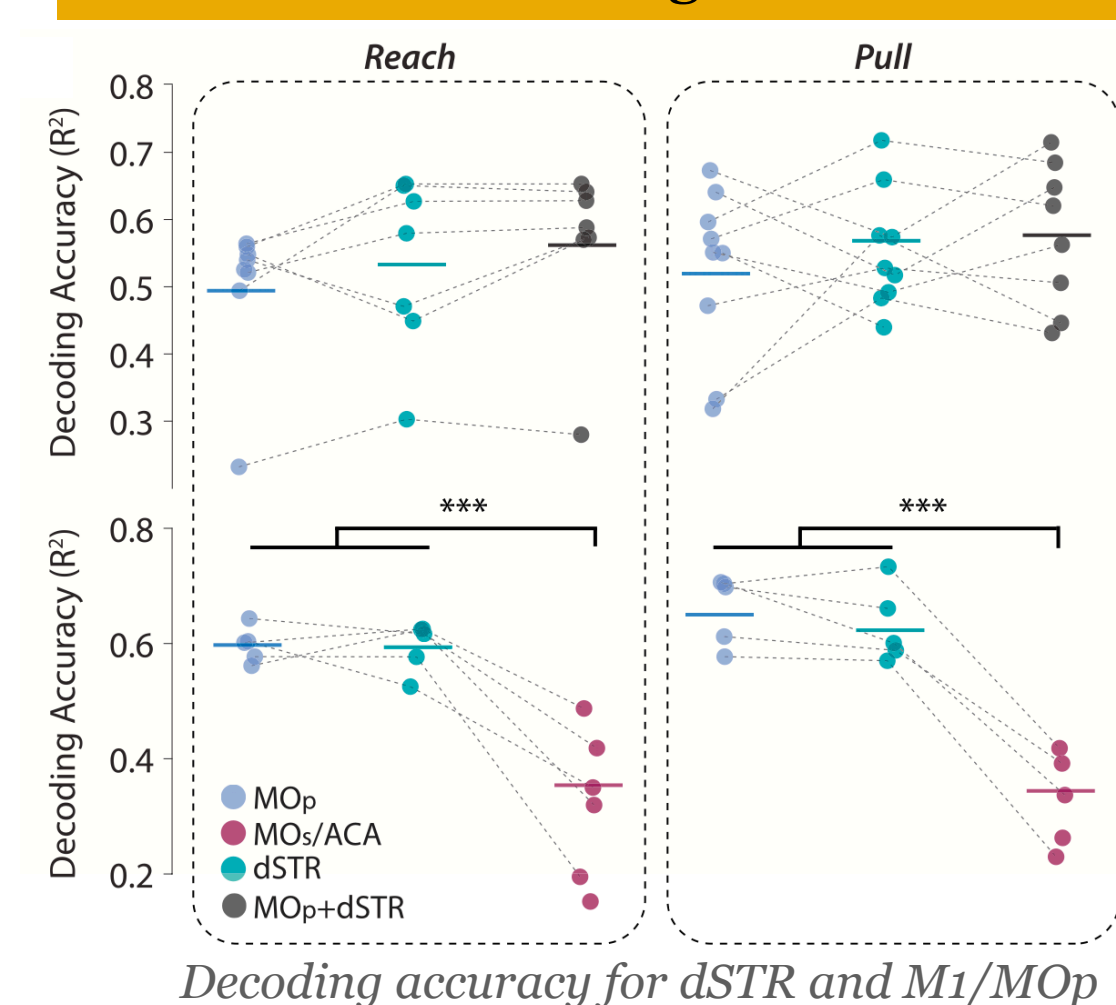


FIXED DISCRETE ACTION



VARIED FORCE

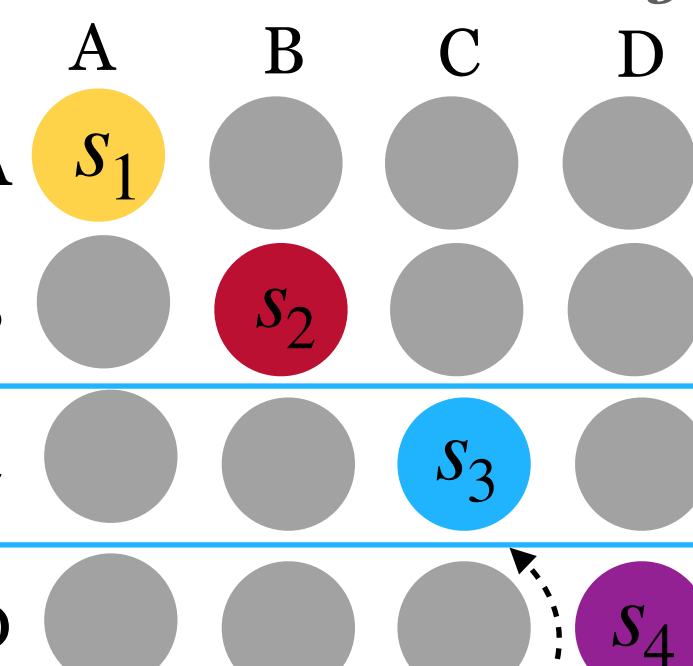
Finding



Decoding accuracy for dSTR and M1/MOP

3 Gap: Current Basal Ganglia Model Limitations

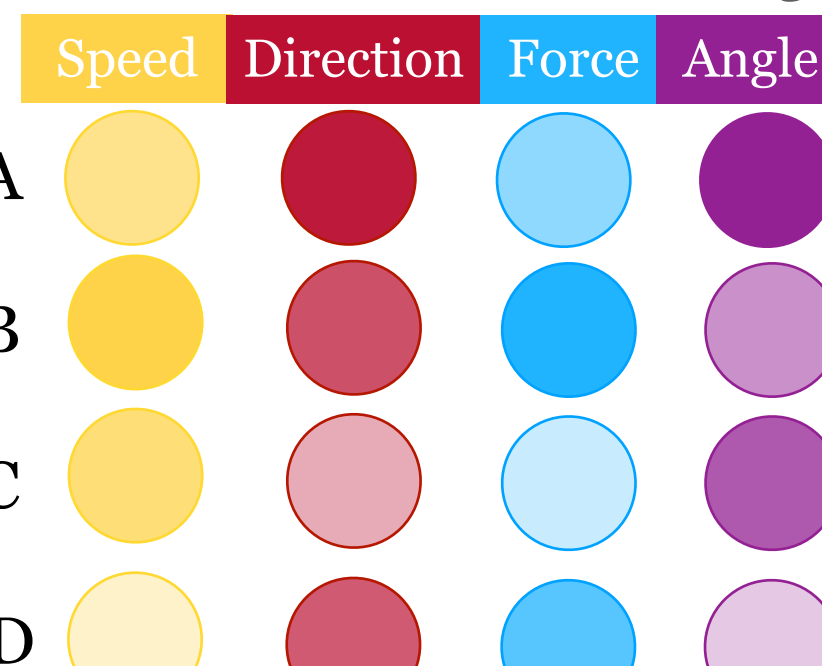
Localist Encoding



Discrete Action Selection

Continuous Action Specification

Distributed Encoding



*Bartlett et al., 2025



Think of distributed encoding as representing attributes of the discrete action.

Why localist models struggle to represent continuous action parameters?

- Binning sacrifices precision
- Brittle: lose a unit, lose a value
- Combinatorial explosion of units (action x continuous parameter)
- Ad-hoc interpolation for intermediate values
- Poor scalability

Bartlett et al. (2025) Model:

Representing a continuous action parameter in a localist framework



Decide which action to perform and how to execute it over multiple discrete action channels

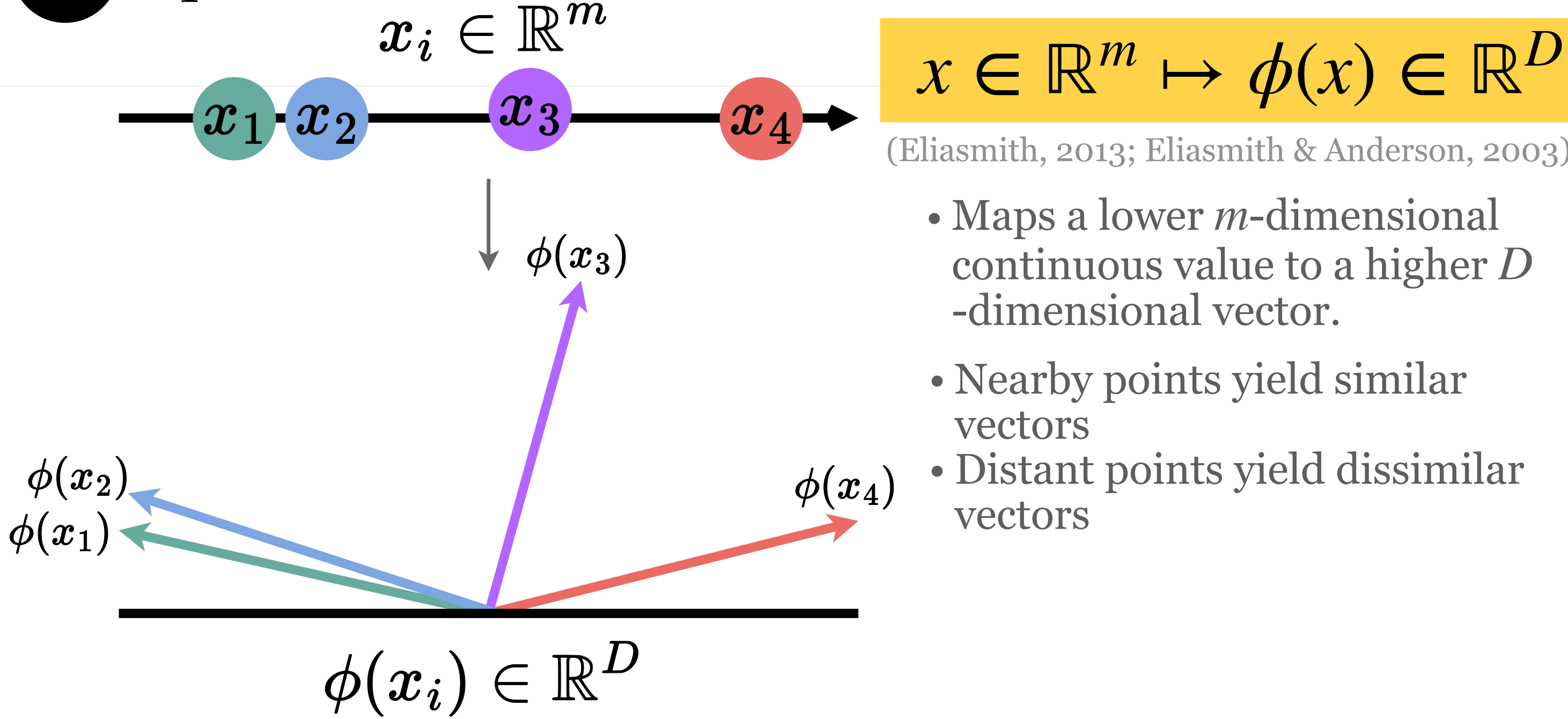
OUR CONTRIBUTION

A single basal ganglia framework that does action selection & action specification in a unified operation over multiple action channels.

Method & Model Ingredients

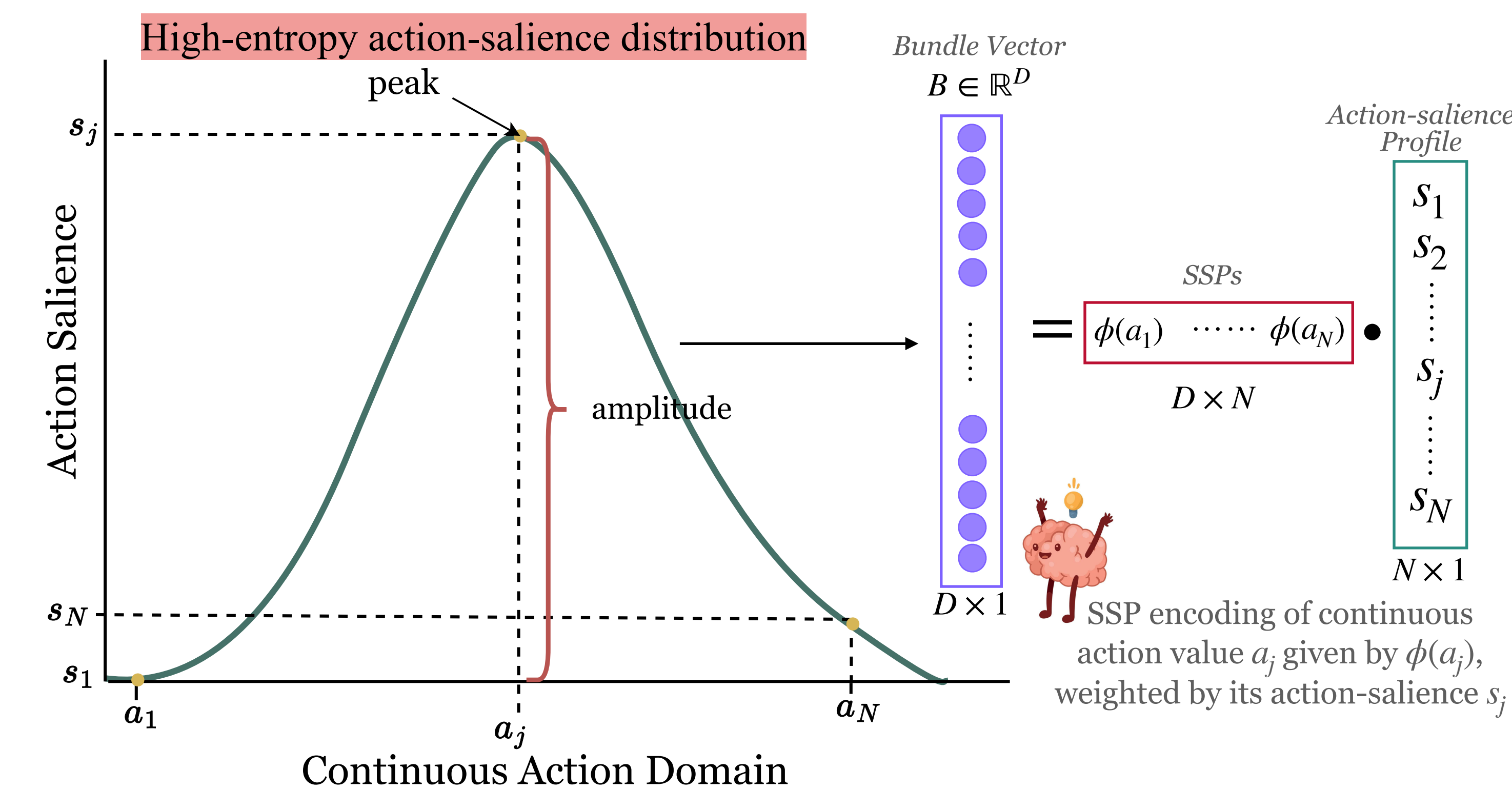
These techniques are adopted from Bartlett et al. (2025) model.

1 Spatial Semantic Pointers (SSPs)



2 Bundling

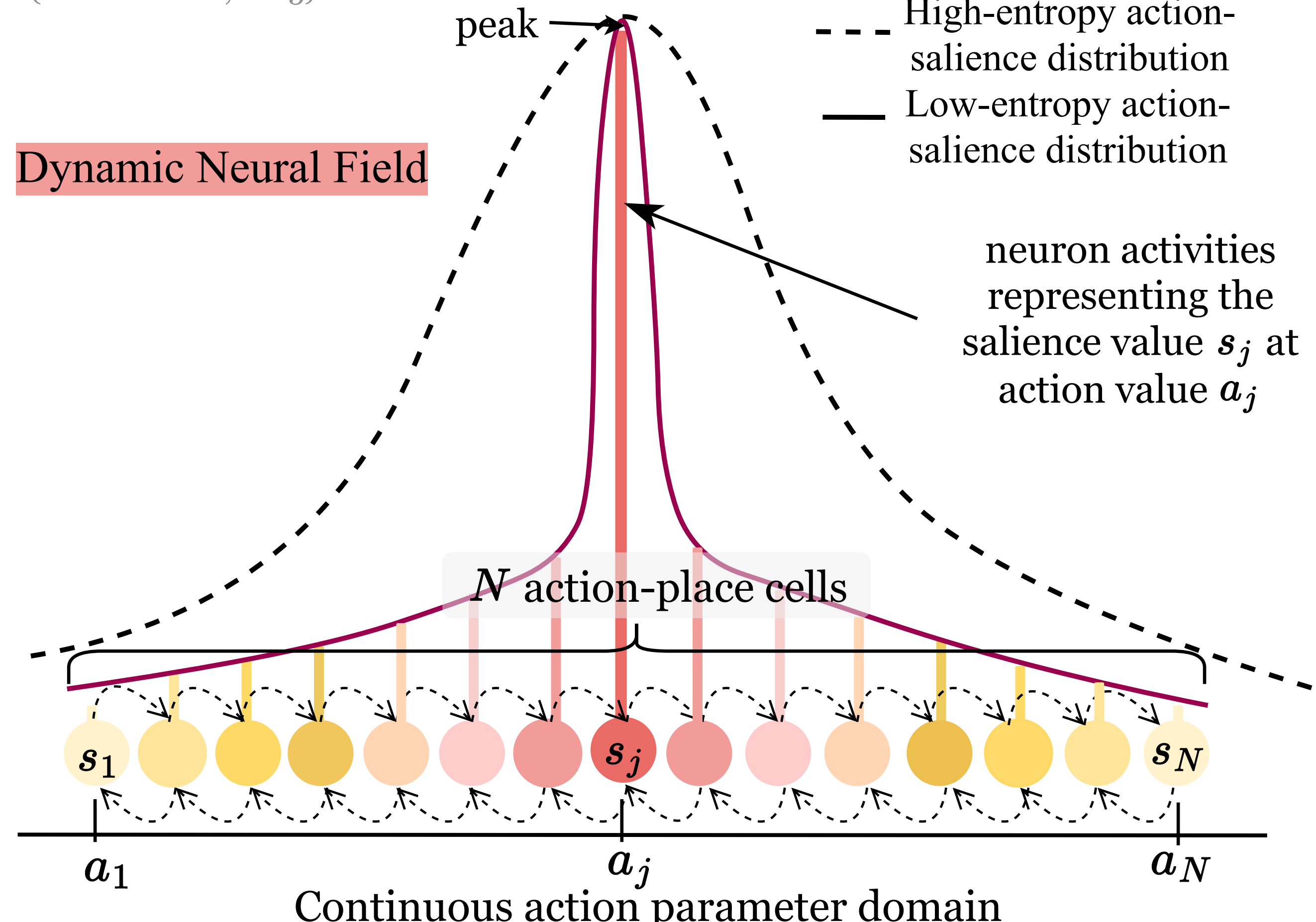
$$\mathbf{B} = s_1\phi(a_1) + \dots + s_N\phi(a_N) = \sum_{j=1}^N s_j\phi(a_j) \in \mathbb{R}^D$$



3 Dynamic Neural Fields

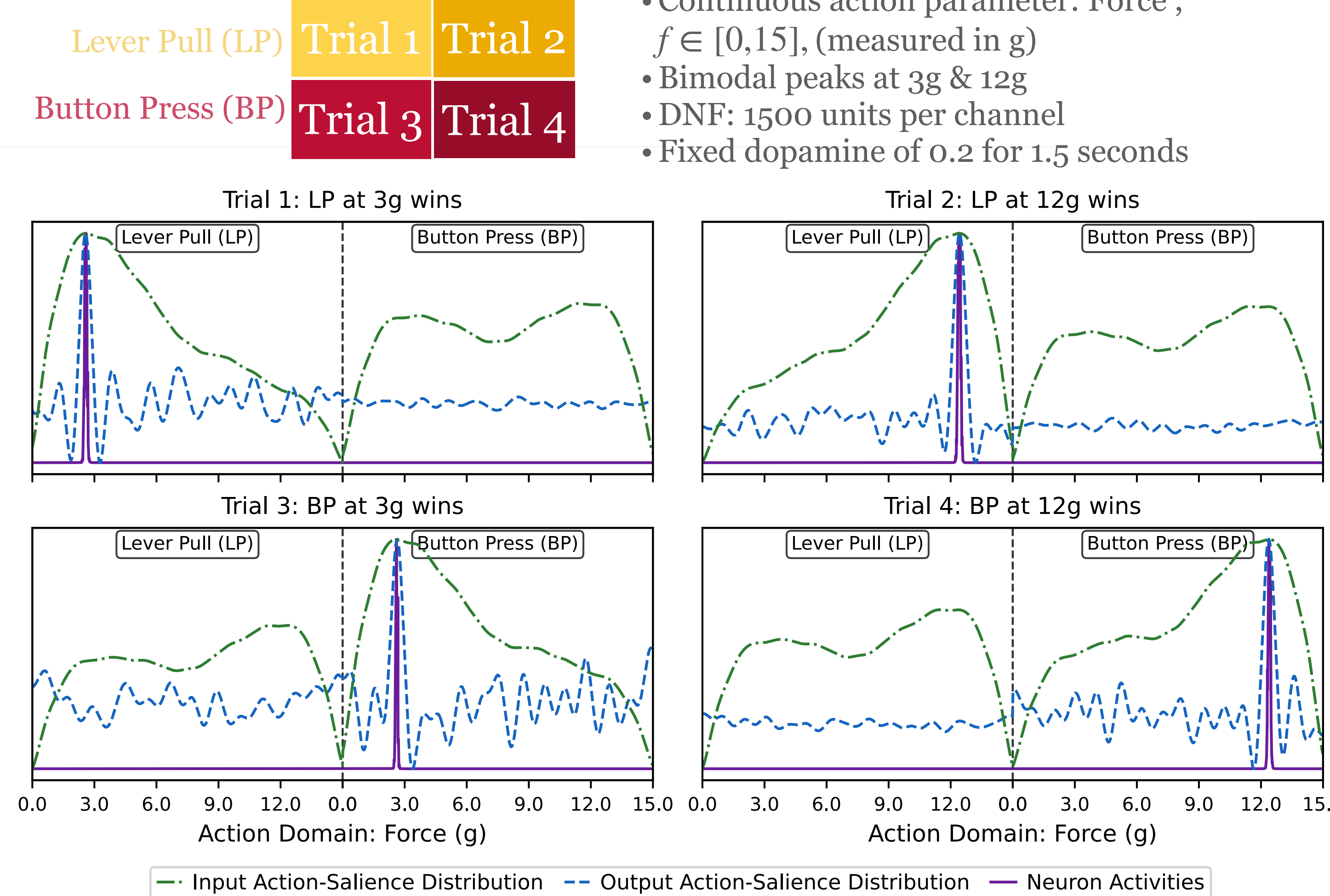
- Entropy-reduction transformation
- Local excitation; global inhibition

(Schöner et al., 2015)



Results & Experiment

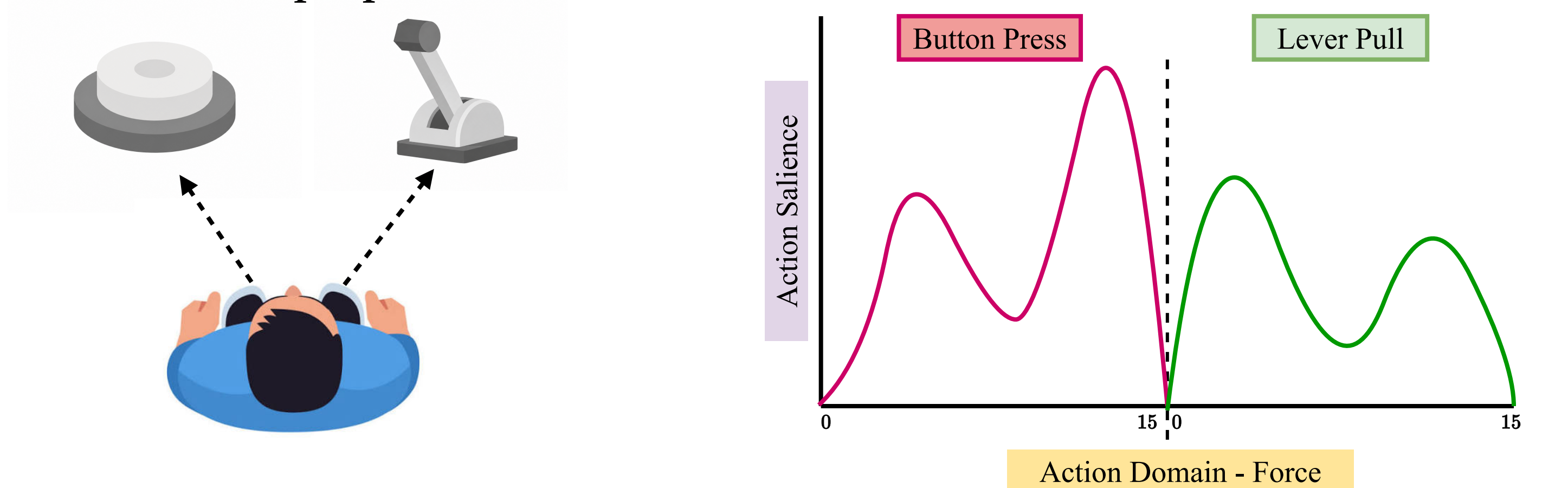
3g 12g



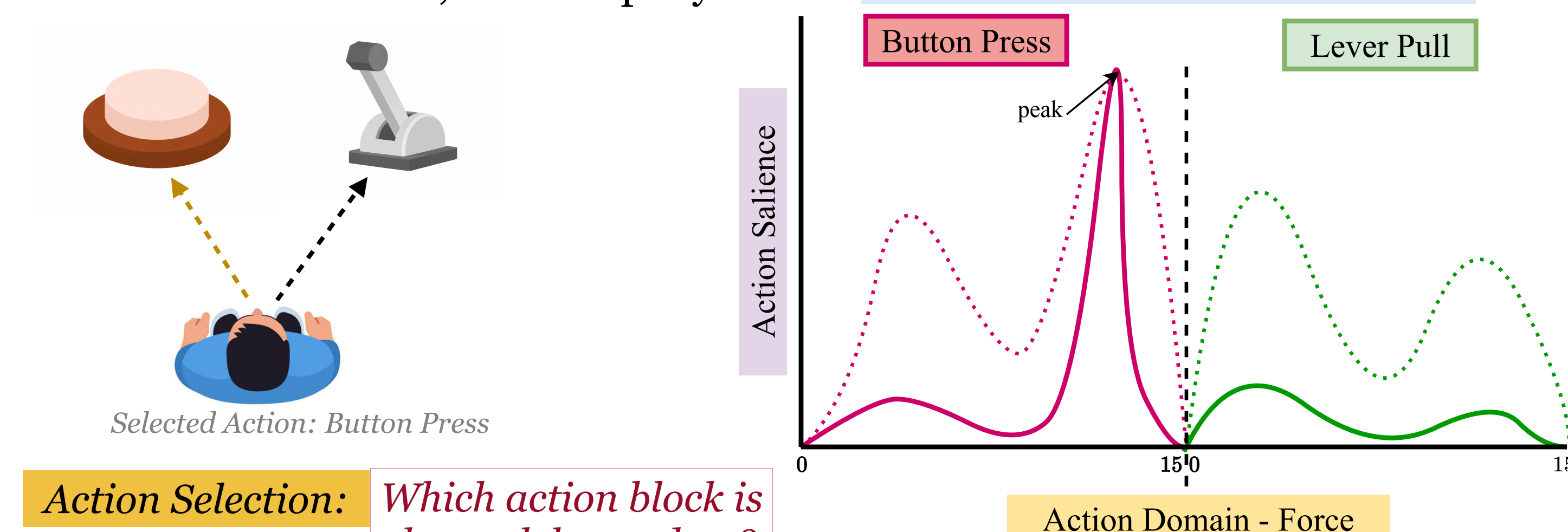
- 2 discrete actions: lever-pull, button-press
- Continuous action parameter: Force, $f \in [0, 15]$, (measured in g)
- Bimodal peaks at 3g & 12g
- DNF: 1500 units per channel
- Fixed dopamine of 0.2 for 1.5 seconds

Putting it all together

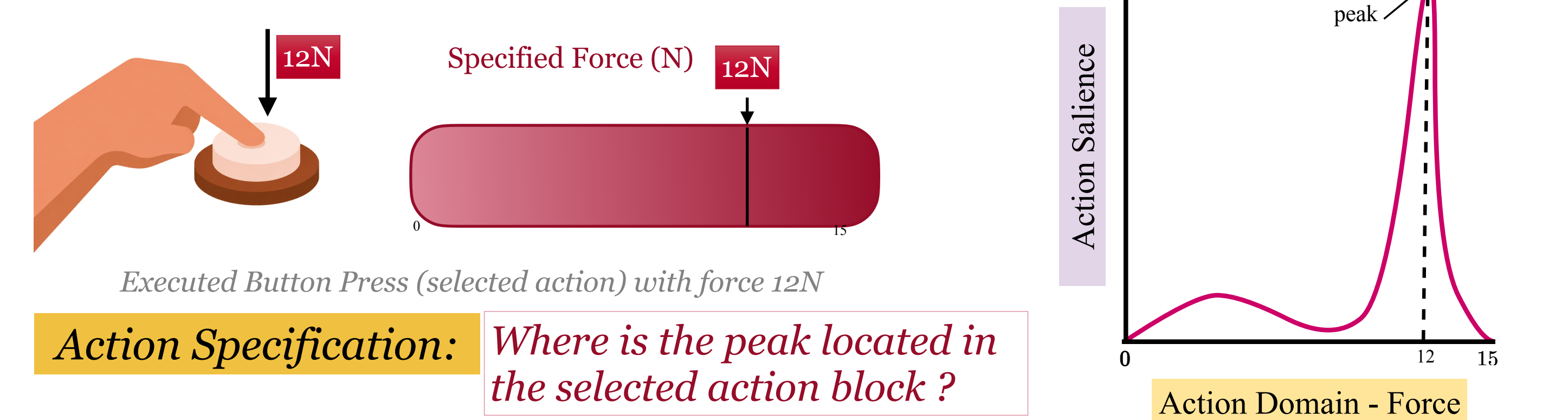
1. Start: Multiple possible actions



2. DNF selects action; “clean-up” system



3. Analogous thalamic-cortical loop chooses the exact execution value



References

- Bartlett, M., Furlong, P. M., Stewart, T. C., & Orchard, J. (2025). A computational model of action specification in the basal ganglia. *bioRxiv*. <https://doi.org/10.1101/2025.08.12.669938>
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- Park, J., Polidoro, P., Fortunato, C., Arnold, J., Mensh, B., Gallego, J. A., & Dudman, J. T. (2025). Conjoint specification of action by neocortex and striatum. *Neuron*, 113, 620–636. <https://doi.org/10.1016/j.neuron.2024.12.024>
- Schöner, G., Spencer, J., & Research Group, D. (2015, November). Dynamic thinking: A primer on dynamic field theory. Oxford University Press. <https://doi.org/10.1093/acprof:oso/978019930563.001.0001>